

Quadratic equations using technology



1 Solve for the pronumeral for each of the following quadratic equations:

a $4m(m + 5) = 0$

b $\frac{m}{2}(m + 5) = 0$

c $-25v^2 + 64 = 0$

d $\frac{x^2}{16} - 2 = 2$

e $81k^2 + 8 = 24$

f $10(p^2 - 7) = 930$

g $4y^2 = 100$

h $25y^2 = 36$

i $-3k^2 = -12$

2 Solve for x for each of the following quadratic equations:

a $(8x + 9)^2 = 256$

b $(x - 4)(x - 2) = 0$

c $x(x + 7) = 0$

d $(8x - 5)(3x - 7) = 0$

e $(3x + 8)(5x - 7) = 0$

f $(3x - 17)(2x + d) = 0$

g $x^2 = 294$

h $81x^2 - 16 = 0$

i $(10x - 9)^2 = 0$

j $(-3 + 7x)^2 = 0$

k $x(2x - 9) = 0$

l $(x - 6)(x + 7) = 0$

m $x^2 = 2$

n $(x - 3)^2 = 64$

o $(x + 3)^2 = 49$

p $(2 - x)^2 = 81$

q $(x - 6)^2 = 2$

r $x^2 = 25$

s $x^2 = 121$

t $x^2 - 121 = 0$

u $x^2 - 10 = 15$

3 Solve the following equations by using technology. Round your answers to one decimal place where necessary.

a $3x^2 - x - 10 = 0$

b $x^2 + 2x - \frac{21}{4} = 0$

c $4.6x^2 + 7.3x - 3.7 = 0$

4 Consider the equation $3x^2 = 6$.

a Solve the equation by using technology, giving your answers in exact form.

b Give the solutions as decimals rounded to the nearest tenth.

5 Consider the equation $0 = -x^2 + 2x - 1$. Use technology to solve the equation for x .



- a Write down how many solutions the equation has.
- b Hence, state how many x -intercepts there are on the graph of the function

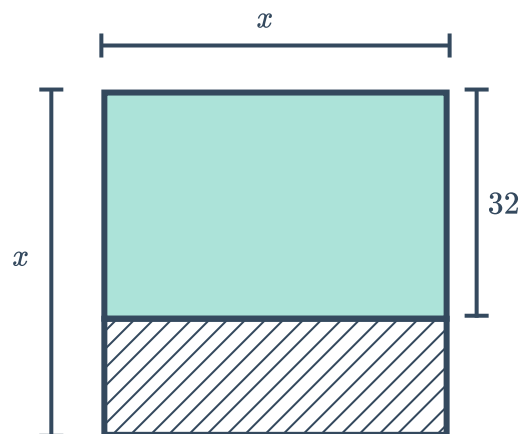
$$y = -x^2 + 2x - 1$$

6 Solve $(5x^2 + 13x + 6)(2x^2 + 13x + 20) = 0$.

Applications

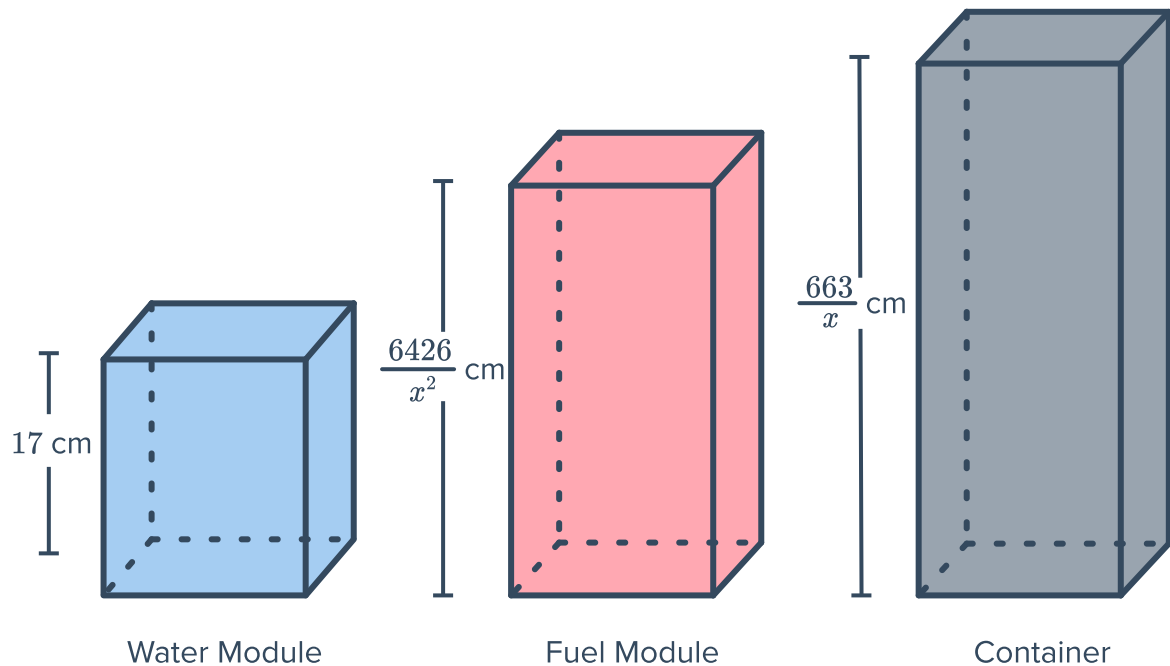
7 The Widget and Trinket Emporium has released the forecast of its revenue over the next year. The revenue R (in dollars) at any point in time t (in months) is described by the equation $R = -(t - 12)^2 + 4$. Solve the equation $-(t - 12)^2 + 4 = 0$ to find the times at which the revenue will be zero.

8 Neville needs a sheet of paper x cm by 13 cm for an origami giraffe. The local origami supply store only sells square sheets of paper. The lower portion of the image shows the excess area A of paper that will be left after Neville cuts out the x cm by 32 cm piece. The excess area, in cm^2 , is given by the equation $A = x(x - 32)$.



- a At what lengths, x , will the excess area be zero?
- b For what value of x will Neville be able to make an origami giraffe with the least amount of excess paper?

- 9 An interplanetary freight transport company has won a contract to supply the space station orbiting Mars. They will be shipping stackable containers, each carrying a fuel module and a water module, that must meet certain dimension restrictions. The design engineers have produced a sketch for the modules and container, shown below. The sum of the heights of both modules equal to the height of the container.



- Write an equation that equates the height of the container and the sum of the heights of the modules.
- Find the possible values of x .
- Find the tallest possible height of the container. Give your answer to two decimal places.